

100 Days of Linux - Day 025

Title: Mini Project #1 - Build and Organize a Project Workspace

Today's Goal

Complete your first real-world Linux mini project by combining everything you've learned so far. No new commands are introduced today.

Why This Matters

Developers and data engineers organize projects, create documentation, search files, inspect logs, and manage folders every day. This project simulates a small part of that workflow.

Commands Used

pwd, ls, cd, mkdir, touch, cp, mv, rm, find, grep, cat, head, tail, nano, wildcards (*, ?, []), less, more

Mini Project Tasks

1. Create a folder named `data_engineering_project`. Inside it create three folders: `python`, `sql`, and `linux`.
2. Using Nano, create `README.md` in the main folder explaining the project in at least 8 lines.
3. Inside each subfolder, create two text files. Add meaningful content using Nano.
4. Use wildcards to list all `.txt` files. Use `find` to locate `README.md`. Use `grep` to search for the word 'Linux' inside all text files.
5. Create a backup folder. Copy every `.txt` file into backup, rename one copied file, verify the results using `ls`, `head`, `tail`, and finally display your command history.

Hints

Complete the project step by step. Verify your work after every major action using `ls`, `cat`, `head`, `tail`, or `find`.

Project Checklist

- ✓ Project folder created
- ✓ README.md completed
- ✓ Six text files created
- ✓ Wildcards used
- ✓ find used
- ✓ grep used
- ✓ Backup folder created
- ✓ Files copied and renamed
- ✓ Verification completed

Common Mistakes

Trying to complete several steps at once. Build the project one

task at a time and verify each step.

Did You Know?

This type of folder organization is similar to how many software and data engineering projects are structured.

Pro Tip

Professional engineers keep README files updated so others can quickly understand a project.

Answer Key (Instructor Only)

Task 1: mkdir data_engineering_project && cd data_engineering_project && mkdir python sql linux

Task 2: nano README.md

Task 3: Create files with nano inside each folder

Task 4: ls *.txt (where applicable), find . -name README.md, grep -r 'Linux' .

Task 5: mkdir backup && cp */*.txt backup/ && mv backup/*.txt backup/.txt && ls backup && head -n 3 backup/ && tail -n 3 backup/ && history

Homework

Repeat this project from scratch in a new folder without using the answer key.

Next Day Preview

Tomorrow you'll continue improving your searching skills with more practical find and grep exercises before moving to permissions.